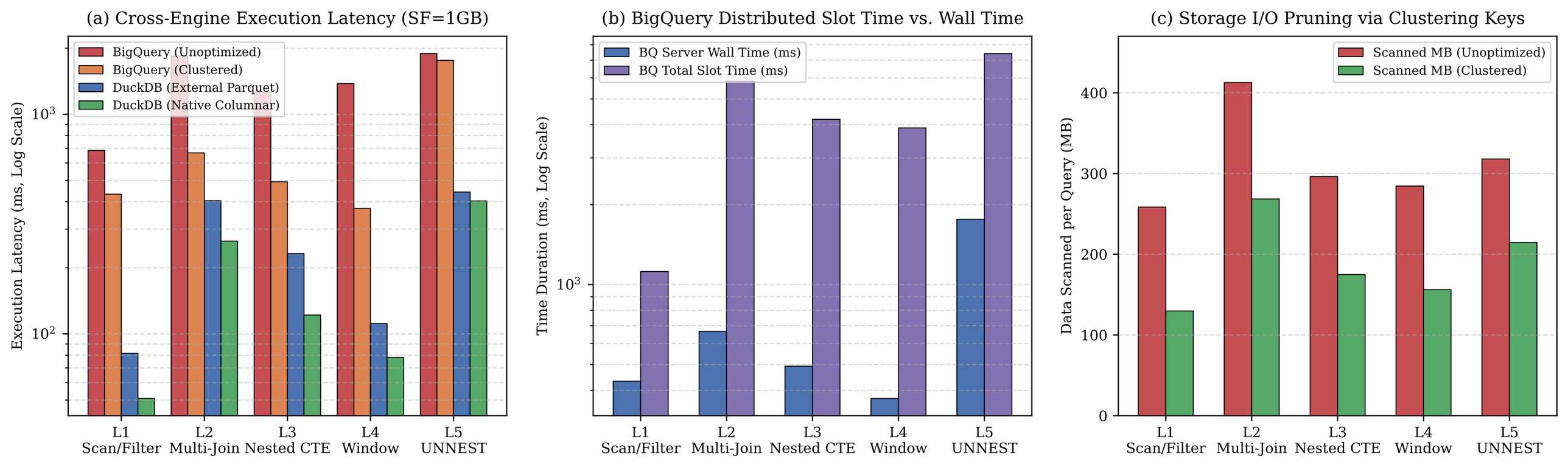
(a) Cross-Engine Execution Latency (SF=1GB)
Execution Latency (ms, Log Scale)
10^3
10^2
BigQuery (Unoptimized)
BigQuery (Clustered)
DuckDB (External Parquet)
DuckDB (Native Columnar)
L1 Scan/Filter
L2 Multi-Join
L3 Nested CTE
L4 Window
L5 UNNEST
(b) BigQuery Distributed Slot Time vs. Wall Time
Time Duration (ms, Log Scale)
10^3
BQ Server Wall Time (ms)
BQ Total Slot Time (ms)
L1 Scan/Filter
L2 Multi-Join
L3 Nested CTE
L4 Window
L5 UNNEST
(c) Storage I/O Pruning via Clustering Keys
Data Scanned per Query (MB)
0
100
200
300
400
Scanned MB (Unoptimized)
Scanned MB (Clustered)
L1 Scan/Filter
L2 Multi-Join
L3 Nested CTE
L4 Window
L5 UNNEST